

DAR ES SALAAM INTL, Tanzania, United Republic ofWMO: **638940**Lat: **6.878S**Lon: **39.203E**Elev: **56**StdP: **100.66**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	19.0	15.6	11.2	26.8	16.8	12.1	25.8	10.2	27.8	9.4	28.2	0.7	200	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	33.5	26.4	33.0	26.1	32.3	25.7	27.6	31.8	27.1	31.1	26.7	30.4	5.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.9	29.8	26.1	21.6	29.5	25.8	21.3	29.1	88.3	32.1	86.0	31.5	84.1	30.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.4	7.9	DB	16.2	35.0	1.7	0.6	15.0	35.5	14.0	35.8	13.1	36.2	11.9	36.6	(4)
(5)			WB	15.1	28.7	1.8	0.7	13.8	29.3	12.8	29.7	11.8	30.1	10.5	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	28.6	28.6	28.3	27.0	26.1	25.1	24.4	24.6	25.2	26.3	27.2	28.3	(6)
(7)		DBStd	1.85	1.09	1.10	1.18	1.16	0.96	0.96	0.93	0.98	0.93	1.09	1.09	1.12	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6073	578	522	566	509	499	453	447	453	456	505	517	568	(10)
(11)		CDD18.3	3032	319	288	308	259	241	203	188	195	206	247	267	310	(11)
(12)		CDH23.3	27982	3608	3268	3233	2159	1758	1383	1267	1379	1654	2188	2627	3457	(12)
(13)		CDH26.7	9864	1339	1286	1260	653	523	407	345	407	566	813	960	1307	(13)
(14)	Wind	WSAvg	3.4	5.1	4.3	2.7	2.6	3.0	3.2	3.5	3.1	3.0	3.1	3.0	4.0	(14)
(15)	Precipitation	PrecAvg	1120	67	59	142	265	167	36	23	26	65	119	114	(15)	
(16)		PrecMax	1733	268	221	426	569	406	154	155	107	96	285	425	377	(16)
(17)		PrecMin	725	1	0	26	13	15	0	1	1	2	0	0	0	(17)
(18)		PrecStd	278	58	52	80	111	100	39	28	24	24	60	108	84	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	34.2	34.2	33.0	32.0	31.2	31.0	31.2	32.0	33.0	33.2	33.8	(19)
(20)			MCWB	26.5	26.5	27.0	26.4	24.9	23.8	23.0	23.2	23.5	24.6	25.3	26.6	(20)
(21)		2%	DB	33.0	33.2	33.2	32.1	31.1	30.4	30.1	30.4	31.1	32.0	32.2	32.9	(21)
(22)			MCWB	26.1	26.2	26.7	26.2	24.8	23.6	22.8	22.9	23.1	24.4	25.2	26.1	(22)
(23)		5%	DB	32.2	32.8	32.8	31.2	30.3	29.9	29.5	29.9	30.5	31.2	31.8	32.1	(23)
(24)			MCWB	25.7	26.0	26.5	25.9	24.6	23.3	22.5	22.6	22.9	24.2	25.2	25.8	(24)
(25)		10%	DB	31.7	32.0	32.0	30.2	29.8	29.1	28.8	29.1	29.9	30.7	31.0	31.6	(25)
(26)			MCWB	25.6	25.7	26.2	25.8	24.5	23.0	22.3	22.4	22.8	24.1	25.1	25.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.9	28.7	27.7	26.5	25.5	24.9	24.8	25.5	26.3	26.7	27.6	(27)
(28)			MADB	32.2	33.1	33.0	31.2	29.8	29.7	29.3	29.7	30.1	30.6	30.8	32.1	(28)
(29)		2%	WB	26.9	27.1	27.7	27.1	26.0	24.8	24.1	24.1	24.6	25.6	26.3	27.0	(29)
(30)			MADB	31.2	31.6	31.8	30.5	29.3	28.7	28.5	28.6	29.2	30.0	30.1	30.9	(30)
(31)		5%	WB	26.5	26.5	27.1	26.6	25.5	24.3	23.5	23.6	24.0	25.1	26.0	26.6	(31)
(32)			MADB	30.3	30.6	30.8	29.8	28.7	28.1	27.8	27.9	28.5	29.4	29.7	30.2	(32)
(33)		10%	WB	26.1	26.2	26.6	26.1	25.1	23.8	23.0	23.0	23.5	24.6	25.6	26.2	(33)
(34)			MADB	29.6	29.9	30.0	29.0	28.2	27.4	27.0	27.2	27.8	28.7	29.2	29.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	7.4	7.9	6.9	7.5	8.9	9.6	9.8	10.2	9.3	8.1	7.1	(35)
(36)			MCDBR	7.3	8.3	8.8	8.3	8.6	9.7	10.4	10.7	10.9	10.1	8.9	7.7	(36)
(37)			MCWBR	2.5	2.9	3.1	3.4	3.4	3.8	4.2	4.2	4.0	3.6	3.0	2.6	(37)
(38)		5% WB	MCDBR	6.7	7.6	8.1	7.5	7.5	8.7	9.7	9.9	10.0	9.1	7.8	7.1	(38)
(39)			MCWBR	2.8	3.0	3.5	3.5	3.5	3.9	4.4	4.4	4.3	3.8	3.1	2.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.480	0.485	0.432	0.401	0.402	0.397	0.404	0.406	0.405	0.407	0.407	0.442	(40)	
(41)		taud	2.248	2.243	2.429	2.529	2.504	2.487	2.432	2.417	2.413	2.425	2.457	2.383	(41)	
(42)		Ebn at Noon	865	860	893	894	866	855	855	878	904	918	925	896	(42)	
(43)		Edh at Noon	147	148	121	105	103	102	110	116	121	122	118	128	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.84	5.99	5.55	4.78	4.59	4.69	4.81	5.14	5.70	6.01	6.04	5.93	(44)	
(45)		RadStd	0.39	0.35	0.38	0.42	0.40	0.19	0.17	0.15	0.20	0.22	0.28	0.33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page